SLM Agentic Benchmark Results

March 5, 2026

Benchmarks: Summarization (XSum) · BIG-bench Lite · PlanBench (Blocksworld)

Models: Azure (OneShotAgent) · Ollama local (OllamaAgent, agentic variants)

Note: Results marked (preliminary) are from in-progress runs. All other results are final ($n \geq 50$ or $n = 100$).

Methods

Summarization (XSum). Each task presents a BBC news article from the XSum validation split. The model returns a JSON object with a `response` field containing a single-sentence summary. Primary metric: ROUGE-L F1 ($\geq$0.15 = success, Wilson 95% CI). Secondary metric: BERTScore F1 (bootstrap 95% CI, 2000 resamples). Two BERTScore models are reported: `distilbert-base-uncased` (used in the Feb 25 report) and the new `microsoft/deberta-xlarge-mnli`, which provides stricter semantic similarity scoring. $n = 100$ per model. Run date: Feb 25, 2026 (Azure); Feb–Mar 2026 (Ollama).

BIG-bench Lite. Six tasks from the BIG-bench benchmark (`tasksource/bigbench` on HuggingFace): logical deduction, navigation, penguins-in-a-table, causal judgment, date understanding, and arithmetic. All tasks are multiple-choice; accuracy = fraction of correct letter choices. One-shot: $n = 16$ per config (96 total, *final*); $n = 50$/config (300 total) run in progress. Agentic runs: $n = 30$/config (180 total) using CrewAI-based Sequential, Concurrent, and GroupChat agent architectures. Run date: Mar 5, 2026.

PlanBench (Blocksworld). Classical AI planning benchmark (Valmeekam et al., 2022; github.com/karthikv792/L Planning). The model is given a description of a set of colored blocks on a table and must produce a sequence of `(unstack A B)` / `(stack A B)` / `(pick-up A)` / `(put-down A)` actions to reach a goal configuration. Success requires *exact* plan correctness verified by a PDDL planner. Two configs: `blocksworld` (4 blocks, harder) and `blocksworld_3` (3 blocks, easier). Ten task types probe different planning competencies: **t1** Plan Generation (generate a valid plan from scratch), **t2** Plan Optimality (produce the shortest plan), **t3** Plan Verification (judge whether a given plan is correct), **t4** Plan Reuse (adapt a stored plan), **t5** Plan Generalization (apply a plan pattern to a new instance), **t6** Replanning (fix a broken plan), **t7** Plan Execution (simulate executing a plan and report final state), **t8_1** Goal Shuffling (reorder goal conditions without changing the underlying plan), **t8_2** Full-to-Partial (identify which subgoals a plan achieves), **t8_3** Partial-to-Full (complete a partially specified plan). $n = 100$ per task/config/model. Run date: Mar 5, 2026.

Confidence intervals. For binary success rates: Wilson score intervals (95%). For BERTScore means: bootstrap percentile intervals (2000 resamples, 95%). CIs wider than ±15pp indicate high uncertainty.

1 Summarization (XSum) — ROUGE-L, all models

TL;DR: Azure frontier models (mistral-small, deepseek-v3.2, ministral-3b) lead at 66–77% ROUGE-L success. Mid-size Ollama models (gemma3-4b, gpt-oss-20b, dasd-4b) reach 52–63%, competitive with several Azure models. Three Azure models (deepseek-r1, deepseek-v3, mistral-nemo) score 0% ROUGE-L due to wording divergence from the reference, not poor summarization quality — their BERTScore confirms semantically valid outputs.

Table 1. Summarization (XSum), $n = 100$ per model. Sorted by ROUGE-L success rate. Wilson 95% CI shown.

Model	n	RL Succ. (%)	Avg RL	RL 95% CI (Wilson)	Provider
mistral-small	100	77.0	0.225	[67.8, 84.2]	Azure
deepseek-v3.2	100	74.0	0.224	[64.6, 81.6]	Azure
ministral-3b	100	70.0	0.208	[60.4, 78.1]	Azure
gpt-4o	100	67.0	0.191	[57.3, 75.4]	Azure
mistral-large-3	100	66.0	0.194	[56.3, 74.5]	Azure
gemma3-4b	100	63.0	0.187	[53.2, 71.8]	Ollama
phi-4	100	59.0	0.177	[49.2, 68.1]	Azure
gemma3n-e2b	100	57.0	0.164	[47.2, 66.3]	Ollama
gpt-oss-20b	100	56.0	0.178	[46.2, 65.3]	Ollama
gemma3-1b	100	55.0	0.176	[45.2, 64.4]	Ollama
gemma3n-e4b	100	55.0	0.178	[45.2, 64.4]	Ollama
dasd-4b	100	52.0	0.164	[42.3, 61.5]	Ollama
llama-3.2-11b-vision	100	51.0	0.160	[41.3, 60.6]	Azure
phi-4-mini	100	48.0	0.157	[38.5, 57.7]	Azure
phi4-mini-reasoning	100	43.0	0.136	[33.7, 52.8]	Ollama
qwen3-0.6b	100	42.0	0.141	[32.8, 51.8]	Ollama
phi-4-mini-reasoning	100	20.0	0.098	[13.3, 28.9]	Azure
llama-3.3-70b	100	19.0	0.075	[12.5, 27.8]	Azure
falcon-h1-90m	100	8.0	0.083	[4.1, 15.0]	Ollama
0% ROUGE-L — semantically valid paraphrases with different wording (BERTScore captures; see Table 2):					
deepseek-r1	100	0.0	0.025	[0.0, 3.7]	Azure
deepseek-v3	100	0.0	0.025	[0.0, 3.7]	Azure
mistral-nemo	100	0.0	0.000	[0.0, 3.7]	Azure

RL = ROUGE-L F1 (threshold $\geq$0.15). falcon-h1-90m is a 90M parameter model. phi4-mini-reasoning = Ollama variant.
Changes from Feb 25: All Ollama models now have $n = 100$ (previously $n = 3$ for most). Azure results unchanged. Ollama models gemma3-1b/4b, gemma3n-e2b/e4b, gpt-oss-20b, qwen3-0.6b, falcon-h1-90m, phi4-mini-reasoning upgraded from $n = 3$ to $n = 100$.

2 Summarization (XSum) — BERTScore: distilbert vs. DeBERTa

TL;DR: DeBERTa-xlarge (new) ranks models more consistently with ROUGE-L than distilbert did. The 5 weakest Ollama models cluster at DeBERTa F1 = 0.349 — they produce nearly identical outputs regardless of input, so the model cannot differentiate them further. gpt-oss-20b's inflated distilbert score (0.817 at

$n = 3$) normalizes to 0.661 at $n = 100$, correctly placing it in the mid tier. Rankings at the top are stable across both models.

Comparison of two BERTScore models. `distilbert-base-uncased` was reported on Feb 25; `microsoft/deberta-x` is the new high-capacity model scored this week. DeBERTa produces lower absolute F1 values but better discriminates models in the mid range, and corrects the inflated score for gpt-oss-20b (previously $n = 3$).

Table 2. BERTScore F1 comparison: distilbert-base-uncased (Feb 25) vs. deberta-xlarge (Mar 5). All runs $n = 100$.

Model	n	Distilbert F1	Distilbert 95% CI	DeBERTa F1	DeBERTa 95% CI	Provider
mistral-small	100	0.800	[0.791, 0.809]	0.702	[0.689, 0.716]	Azure
deepseek-v3.2	100	0.799	[0.791, 0.808]	0.690	[0.677, 0.702]	Azure
ministral-3b	100	0.789	[0.780, 0.797]	0.682	[0.670, 0.695]	Azure
mistral-large-3	100	0.792	[0.784, 0.801]	0.678	[0.665, 0.691]	Azure
gpt-4o	100	0.790	[0.783, 0.798]	0.675	[0.663, 0.687]	Azure
phi-4	100	0.784	[0.776, 0.791]	0.667	[0.655, 0.678]	Azure
gpt-oss-20b	100	0.661†	[0.649, 0.673]	0.661	[0.649, 0.673]	Ollama
dasd-4b	100	0.779	[0.772, 0.787]	0.649	[0.637, 0.661]	Ollama
phi-4-mini	100	0.770	[0.760, 0.781]	0.648	[0.631, 0.664]	Azure
qwen3-0.6b	100	0.638†	[0.629, 0.647]	0.638	[0.629, 0.647]	Ollama
llama-3.2-11b-vision	100	0.748	[0.732, 0.764]	0.611	[0.584, 0.637]	Azure
phi-4-mini-reasoning	100	0.704	[0.693, 0.715]	0.554	[0.540, 0.569]	Azure
llama-3.3-70b	100	0.682	[0.669, 0.696]	0.489	[0.467, 0.513]	Azure
deepseek-r1	100	0.649	[0.644, 0.654]	0.439	[0.435, 0.443]	Azure
deepseek-v3	100	0.649	[0.644, 0.654]	0.439	[0.436, 0.443]	Azure
mistral-nemo	100	0.647	[0.643, 0.652]	0.411	[0.407, 0.416]	Azure
gemma3-1b	100	0.578†	[0.535, 0.614]	0.349	[0.345, 0.353]	Ollama
gemma3-4b	100	0.578†	[0.535, 0.614]	0.349	[0.345, 0.353]	Ollama
gemma3n-e2b	100	0.578†	[0.535, 0.614]	0.349	[0.345, 0.353]	Ollama
gemma3n-e4b	100	0.578†	[0.535, 0.614]	0.349	[0.345, 0.353]	Ollama
falcon-h1-90m	100	0.578†	[0.535, 0.614]	0.349	[0.345, 0.353]	Ollama
phi4-mini-reasoning	100	0.578†	[0.535, 0.614]	0.349	[0.345, 0.353]	Ollama

† These models had $n = 3$ in the Feb 25 distilbert run. Their distilbert scores here are from the new $n = 100$ run.
Key finding: DeBERTa-xlarge more sharply penalizes format/style divergence. The 5 lowest-ranked Ollama models (gemma3-1b/4b, gemma3n, falcon, phi4-mini-reasoning) cluster at 0.349 DeBERTa F1 vs. 0.578 distilbert – a sign these models produce repetitive or near-identical outputs.

deepseek-r1/v3 and mistral-nemo: 0% ROUGE-L but non-trivial distilbert BERTScore (≈0.648); DeBERTa collapses them to 0.41–0.44, correctly reflecting their lexically divergent outputs.

3 BIG-bench Lite — One-Shot

TL;DR: gpt-oss-20b (74%) outperforms dasd-4b (59.4%) overall. Both models are near-perfect on arithmetic and penguins-in-table (structured lookup tasks) but struggle on navigation (50%/31%) and causal judgment (50%/44%). Wide per-task CIs at $n = 16$/config should narrow significantly in the $n = 50$/config run currently in progress.

Six-task subset of BIG-bench (`logical_deduction, navigate, penguins_in_a_table, causal_judgment, date_understanding, arithmetic`). All multiple-choice; accuracy = fraction correct. $n = 16$ per config (96 total) is **final**. A $n = 50$/config (300 total) run is in progress at time of writing.

Table 3. BIG-bench Lite one-shot, $n = 96$ (16/config). Wilson 95% CIs. **Final.**

Model	n	Accuracy (%)	95% CI (Wilson)	Provider
gpt-oss-20b	96	74.0	[64.4, 81.7]	Ollama
dasd-4b	96	59.4	[49.4, 68.7]	Ollama

Table 4. BIG-bench Lite one-shot per-task breakdown, $n = 16$/config. (n=300 run in progress – preliminary figures in parentheses for gpt-oss-20b).

Task	gpt-oss-20b		dasd-4b	
	Acc (%)	95% CI	Acc (%)	95% CI
logical_deduction	62	[38.6, 81.5]	25	[10.2, 49.5]
navigate	50	[28.0, 72.0]	31	[14.2, 55.6]
penguins_in_table	100	[80.6, 100]	94	[71.7, 98.9]
causal_judgment	50	[28.0, 72.0]	44	[23.1, 66.8]
date_understanding	81	[57.0, 93.4]	62	[38.6, 81.5]
arithmetic	100	[80.6, 100]	100	[80.6, 100]
Overall	**74.0**	[64.4, 81.7]	**59.4**	[49.4, 68.7]

Wide CIs on $n = 16$/task are expected. The $n = 50$/task run will narrow CIs to $\approx \pm 13$pp. Both models score 100% on arithmetic, indicating strong symbolic reasoning at this difficulty level.

4 BIG-bench Lite — Agentic Architectures (gpt-oss-20b)

TL;DR: Both agentic architectures substantially outperform one-shot. Sequential (+12.6pp) and preliminary Concurrent (+14.6pp) show the largest gains on reasoning-heavy tasks (logical deduction +31pp, navigation +33pp) — the multi-role pipeline effectively forces the model to re-examine its reasoning before committing. GroupChat and all dasd-4b agentic results are still pending.

Three agentic architectures implemented with CrewAI + Ollama LiteLLM backend. Each uses a multi-role pipeline calling gpt-oss-20b for all roles:

- **Sequential:** Analyzer → Evaluator → Responder (3 serial LLM calls per task)
- **Concurrent:** Analyst + Researcher + Critic (parallel) → Synthesizer (4 calls total)
- **GroupChat:** Proposer → Critic → Advisor → Moderator (4 serial calls)

$n = 30$/config (180 total) per architecture.

Table 5. BIG-bench Lite — gpt-oss-20b agentic results ($n = 180$, 30/config).

Architecture	n	Accuracy (%)	95% CI (Wilson)	Status
One-shot (baseline)	96	74.0	[64.4, 81.7]	Final
Sequential	180	86.6	[80.9, 90.9]	Final
Concurrent	149	88.6	[82.5, 92.8]	Preliminary (149/180)
GroupChat	—	—	—	Pending

One-shot baseline uses $n = 96$ (16/config) run for comparability. Sequential and concurrent show substantial gains over one-shot (+12.6pp and +14.6pp respectively, preliminary for concurrent). GroupChat results pending.

Table 6. Sequential vs. one-shot per-task breakdown (gpt-oss-20b, $n = 30$/config).

Task	One-Shot ($n = 16$)		Sequential ($n = 30$)	
	Acc (%)	95% CI	Acc (%)	95% CI
logical_deduction	62	[38.6, 81.5]	93	[78.0, 98.1]
navigate	50	[28.0, 72.0]	83	[66.4, 92.7]
penguins_in_table	100	[80.6, 100]	90	[74.4, 96.5]
causal_judgment	50	[28.0, 72.0]	70	[52.1, 83.3]
date_understanding	81	[57.0, 93.4]	87	[70.3, 94.7]
arithmetic	100	[80.6, 100]	97	[83.3, 99.4]
Overall	**74.0**	[64.4, 81.7]	**86.6**	[80.9, 90.9]

Sequential shows the largest gains on logical deduction (+31pp) and navigation (+33pp) – tasks requiring step-by-step reasoning that benefits from the Analyzer→Evaluator pipeline. Penguins-in-table and arithmetic were already near-ceiling and show slight regression within CI noise. Causal judgment shows a modest +20pp improvement. dasd-4b agentic results pending.

5 PlanBench (Blocksworld) — All Models, $n = 100$

TL;DR: No model can generate or optimize plans (t1–t2 near 0% universally). All models — even the 90M-param falcon-h1-90m — score highly on robustness tasks (t8_x: 59–100%), which only require pattern-matching on an existing plan rather than generating one. qwen3-0.6b's 97% on plan verification (t3) is a template artifact that collapses to 1% on a different config. The gap between generation (0%) and robustness (90%+) is the defining characteristic of current SLMs on planning.

Classical AI planning on blocksworld (4 blocks) and blocksworld_3 (3 blocks) configurations. Ten task types assess different aspects of planning competence. Success = exact plan match. $n = 100$ per task per model.

Task type legend: t1=Plan Generation, t2=Plan Optimality, t3=Plan Verification, t4=Plan Reuse, t5=Plan Generalization, t6=Replanning, t7=Plan Execution, t8_1=Goal Shuffling, t8_2=Full-to-Partial, t8_3=Partial-to-Full.

Table 7. PlanBench — `blocksworld` (4 blocks), $n = 100$/task. Wilson 95% CI.

Model	t1	t2	t3	t4	t5	t6	t7	t8_1	t8_2	t8_3
qwen3-0.6b	0	0	97	14	1	6	0	97	96	93
dasd-4b	0	0	0	10	8	6	0	94	61	92
gemma3-1b	0	0	–	11	2	7	0	59	73	51
falcon-h1-90m	0	0	0	13	0	7	0	87	–	–

Values are % success. "–" = not run or not evaluated. t3 not run for gemma3-1b in blocksworld config.

Table 8. PlanBench — `blocksworld_3` (3 blocks), $n = 100$/task. Wilson 95% CI.

Model	t1	t2	t3	t4	t5	t6	t7	t8_1	t8_2	t8_3
qwen3-0.6b	0	0	1	17	1	–	5	100	100	93
dasd-4b	5	0	3	15	8	9	0	90	90	95
falcon-h1-90m	–	–	–	–	–	–	–	–	–	–

qwen3-0.6b: t3 (plan verification) scored 97% on blocksworld but only 1% on blocksworld_3 — this is the well-known template-matching artifact: the model learns the fixed verification answer for one scenario. "–" = not run.

Table 9. PlanBench 95% Wilson CIs for selected high-performing (t8_x) and zero-scoring (t1, t2) tasks.

Model	Config	t8_1 Goal Shuffling		t1 Plan Generation	
		Acc (%)	95% CI	Acc (%)	95% CI
qwen3-0.6b	blocksworld	97.0	[91.5, 99.0]	0.0	[0.0, 3.7]
dasd-4b	blocksworld	94.0	[87.5, 97.2]	0.0	[0.0, 3.7]
gemma3-1b	blocksworld	59.0	[49.2, 68.1]	0.0	[0.0, 3.7]
falcon-h1-90m	blocksworld	87.0	[79.0, 92.2]	0.0	[0.0, 3.7]
dasd-4b	blocksworld_3	90.0	[82.6, 94.5]	5.0	[2.2, 11.2]
qwen3-0.6b	blocksworld_3	100.0	[96.3, 100]	0.0	[0.0, 3.7]

Key observations:

- **Plan generation (t1, t2) is universally hard.** All models score near 0% on plan generation and optimality tasks, regardless of model size (0.6B to 20B parameters). This aligns with published findings that SLMs cannot perform multi-step classical planning without specialized training.
- **Robustness tasks (t8_x) are surprisingly easy.** Goal shuffling, full-to-partial, and partial-to-full tasks score 59–100% across all models. These tasks require identifying properties of a given plan rather than generating one, which is an easier matching problem.
- **qwen3-0.6b achieves 97% on plan verification (t3) in blocksworld but 1% in blocksworld_3.** This is a template artifact, not generalized reasoning.
- **dasd-4b (4B params) outperforms qwen3-0.6b (0.6B) on plan reuse and generalization**, but qwen3-0.6b edges ahead on the t8 robustness tasks.

6 Cross-Benchmark Summary

Table 10. Cross-benchmark comparison. Summarization = ROUGE-L success (%); BIG-bench = one-shot accuracy (%); Planning = composite success (%) from Feb 25; PlanBench = t8_1 goal shuffling success (%) as representative task.

Model	Summ. RL	BERTScore (DeBERTa)	BIG-bench (OS)	Planning (2/25)	PlanBench t8_1	Provider
mistral-small	77.0	0.702	—	0.0†	—	Azure
deepseek-v3.2	74.0	0.690	—	—	—	Azure
ministral-3b	70.0	0.682	—	42.0	—	Azure
gpt-4o	67.0	0.675	—	80.0	—	Azure
mistral-large-3	66.0	0.678	—	46.0	—	Azure
phi-4	59.0	0.667	—	0.0†	—	Azure
gpt-oss-20b	56.0	0.661	74.0	52.0	—	Ollama
gemma3-4b	63.0	0.349	—	0.0†	—	Ollama
dasd-4b	52.0	0.649	59.4	36.0	94.0	Ollama
qwen3-0.6b	42.0	0.638	—	64.0	97.0	Ollama
gemma3-1b	55.0	0.349	—	0.0†	59.0	Ollama
phi-4-mini	48.0	0.648	—	10.0	—	Azure
phi-4-mini-reasoning	20.0	0.554	—	18.0	—	Azure
llama-3.3-70b	19.0	0.489	—	4.0	—	Azure
llama-3.2-11b-vision	51.0	0.611	—	0.0†	—	Azure
falcon-h1-90m	8.0	0.349	—	—	87.0	Ollama
deepseek-r1	0.0	0.439	—	—	—	Azure
deepseek-v3	0.0	0.439	—	—	—	Azure
mistral-nemo	0.0	0.411	—	0.0†	—	Azure

† = 0% Planning success (output format errors). "—" = benchmark not run for this model.
BIG-bench one-shot: only gpt-oss-20b and dasd-4b have been run to date. Planning results are from Feb 25 (DeepPlanning, $n = 50$). PlanBench t8_1 = Goal Shuffling on `blocksworld` config.

Notable: falcon-h1-90m (90M params) achieves 87% on PlanBench t8_1 but only 8% ROUGE-L on summarization, illustrating that task-specific "planning" robustness does not generalize to free-text generation.

Pending / In-Progress

- **BIG-bench one-shot $n = 50$/config (300 total):** gpt-oss-20b and dasd-4b runs in progress (PID 50357). ETA ≈1h from time of writing.
- **BIG-bench agentic concurrent (gpt-oss-20b):** 149/180 complete at time of writing (≈88.6% accuracy). ETA ≈30 min.
- **BIG-bench agentic GroupChat (gpt-oss-20b):** Pending; will begin after concurrent completes.
- **BIG-bench agentic (dasd-4b):** Sequential, Concurrent, GroupChat all pending; will begin after gpt-oss-20b agentic runs finish.
- **BIG-bench one-shot for additional models:** Only gpt-oss-20b and dasd-4b have been run. Azure models and other Ollama models pending.

- **PlanBench for gpt-oss-20b:** gpt-oss-20b planbench tasks returned 0 tasks evaluated in the Mar 4–5 runs; under investigation.